When Do Relations Help Collective Forecasts? A Controlled Study of Operational Temporal Graphs

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Abstract

Individual event probabilities do not determine uncertainty in group counts or simultaneous delays. We study when observed relations improve collective forecasts across six operational logs spanning compute clusters, cloud VMs, loan workflows, and IT incidents. A current-cohort benchmark separates three effects: stronger individual predictions, shared uncertainty, and typed dependence beyond a global factor. Controls include temporal and graph neural predictors, feature-matched boosted trees, Gaussian and Student-t copulas, direct count distributions, and calibration. Three additional source periods are evaluated under a frozen protocol after small, storage-bounded pilots. On the original cluster trace, typed factors improve count CRPS by 24.3% over independence but only 2.2% over a global factor. On new-period IT incidents, typed Gaussian and Student-t factors improve their global controls by 3.3% and 3.6%, respectively, while a strong bias-adjusted direct-count model is essentially tied with the latter. In Azure, a development-selected subscription-only factor improves global CRPS by 4.7%, although the combined typed fit collapses to the global control. Other sources show smaller incremental gains; larger neural models do not consistently beat trees. Count accuracy, calibration, and reserve-loss rankings also differ. The contribution is a reproducible cross-domain evaluation and controlled evidence about when relations matter, not a universally superior graph architecture or a claim of deployment savings.

1 Introduction

A cluster operator sees tasks belonging to jobs and machines; an incident manager sees unresolved tickets linked to handlers and teams. Both need collective forecasts: how many current entities will remain unresolved, and which groups might fall behind together? Temporal models can supply each entity's event probability, but those probabilities alone do not specify the distribution of group counts or the largest group delay. Moreover, a well-scored count distribution need not support the particular decision quantile used by an operator.

This study asks a practical evaluation question: *what additional evidence is needed before individual forecasts on a temporal operational graph can be used as collective scenarios?* Three issues are easily conflated. Individual predictions may share a mean bias. Outcomes may retain common variation after conditioning on observed history. Typed relations may identify additional association beyond a global factor. Improving the first issue, adding the second, and exploiting the third require different controls. An apparent "graph benefit" against independent sampling can otherwise be mostly a benefit of shared uncertainty available without typed relations.

We define a common current-cohort target for six public logs: a compute-cluster trace, Azure VMs, two loan-workflow releases, and two IT-incident sources. At each origin, a timestamp-bounded adapter reconstructs the active entities and their observed memberships. Scenarios assign each entity a first-exit bin across four horizons, including survival beyond the final horizon. The cohort is not selected using future exits, arrivals or assignments. The resulting counts can be projected onto overlapping jobs and machines or onto recent-handler groups.

The contribution is a reproducible benchmark and controlled empirical study, not a new copula architecture. We compare strong individual predictors, matched-marginal global and typed factors, direct count distributions, historical residuals, and discrete probabilistic reconciliation. Delayed calibration controls separate mean error from dependence, and threshold-free joint scores test whether count accuracy extends to a group-risk profile. An initial two-log evaluation motivates retrospective diagnostics; a separately frozen third-log period and three further frozen source periods supply confirmation. Storage-bounded transportation pilots and incomplete-history audits are retained as negative or deferred evidence. The manuscript identifies these stages explicitly rather than pooling development and confirmation.

Three findings organize the study. Shared factors are strong controls, so the incremental contribution of typed incidence must be measured against them. Job grouping and IT handler/team incidence add predictive signal in particular settings, but graph architecture, relation identity, and dependence are not interchangeable sources of improvement. Calibration can materially change both collective scores and asymmetric loss, but transferring residuals across predictor versions is unreliable. Finally, methods that forecast individual group counts well can rank differently on the largest standardized group delay. These findings motivate an evaluation workflow that

checks marginal bias, group-count uncertainty, and joint group risk separately, rather than treating one score as validation of a scenario generator.

2 Related work

Temporal graphs and event generation. GraphMixer illustrates strong temporal encoding with simple components [Cong et al., 2023]; our Mixer-style survival predictor is an independent adaptation, not a reproduction of its link-prediction protocol. TGN, TGAT, and DyGFormer model temporal graph interactions [Rossi et al., 2020, Xu et al., 2020, Yu et al., 2023]; our stronger controls use typed aggregation and observed-history attention with a current-cohort first-exit head, not their original link predictors. CEP3 models community event times [Wang et al., 2022], and DG-Gen generates continuous-time dynamic graphs [Hosseini et al., 2025]. Their generation targets differ from conditional first exits of an already active population. We make no numerical superiority claim over these systems. Such a comparison requires explicit state initialization, unknown future-count stopping and outcome adapters.

Joint forecasting and reconciliation. TACTiS and TACTiS-2 learn flexible multivariate copulas [Drouin et al., 2022, Ashok et al., 2024]; separating learned marginals from spatial copulas is also established [Huk et al., 2025]. A bounded development pilot adapts the published TACTiS copula module, with explicit discrete-target and rank adjustments; it is not a full TACTiS/TACTiS-2 reproduction. Our positive-factor Gaussian and Marshall–Olkin controls [Marshall and Olkin, 1967] are deliberately restrictive classical families. Typed current incidence specifies shared randomness for changing entities; neither that construction nor the marginal–dependence decomposition is claimed as a new statistical principle.

Probabilistic reconciliation already conditions forecast distributions on aggregation constraints, including discrete counts [Zambon et al., 2024]. We adapt a two-level importance-conditioning step to vectors of first-exit category counts and compare it with direct Dirichlet–multinomial forecasts and bottom-up aggregation. This is an established-method comparator, not a new reconciliation algorithm or an exact reproduction of an entire software package. Proper multivariate scores complement marginal scores; the limited dependence sensitivity of energy scores motivates an additional variogram score [Scheuerer and Hamill, 2015].

Operational forecasting. RIMS combines learned predictions and intercase simulation [Meneghello et al., 2025], while AgentSimulator models resource interactions [Kirchdorfer et al., 2025]. Closely related work initializes business-process simulation from current states and estimates conditional uncertainty, including on BPI 2017 [Avramenko et al., 2026]. Current-state forecasting and operational uncertainty are therefore established problems. Our empirical distinction is to hold individual probabilities fixed when testing incidence, then separately test correction of those probabilities, direct count alternatives and joint group-risk scores across workflow and compute logs. Executable workload generation, such as Mimesys [Kim et al., 2026], has a different output contract: our first-exit scenarios do not reproduce full workloads, future arrivals or repeated process transitions.

3 Forecast contract and controls

Let V_t be the entities active after processing all events recorded by origin t, and let $g_r(i,t)$ denote entity i's observed membership in relation type r. For ordered horizons $h_1 < \cdots < h_K$, the outcome $Y_{ik} = 1\{T_i \le t + h_k\}$ records the first exit from the current task state, VM deletion, or first qualifying workflow/incident milestone. Each forecast draw assigns one of $K+1$ categories, with the final category indicating survival beyond h_K. This contract does not predict subsequent returns, exact within-bin times, or outcomes of future arrivals.

An individual predictor supplies monotone CDFs F_{ik}. For a current group A, the expected exit count is $\sum_{i\in A} F_{ik}$ under every coupling, whereas its variance also includes pairwise outcome covariances. Thus a coupling can change aggregate dispersion and tails but cannot fix a biased expected count. We exploit this distinction in two separate experiments: matched-marginal dependence comparisons and controlled marginal correction.

For a fixed horizon this distinction is explicit:

$$\mathrm{Var}(C_A) = \sum_{i\in A} F_i(1-F_i) + 2 \sum_{i<j;\, i,j\in A} \mathrm{Cov}(Y_i, Y_j). \quad (1)$$

Relations can improve collective uncertainty when they identify residual co-movement relevant to the requested groups. They need not improve individual probabilities, and a relation correlated with outcomes before conditioning need not add information after strong shared-context predictors. This familiar identity motivates the separate marginal, global, typed, and incidence-shuffle controls; it is not a new theoretical claim.

3.1 Entity-coherent factor scenarios

For each draw, generate independent standard normal global, relation-group, and private factors and set

$$Z_i = a_i Z_0 + \sum_r b_{ir} Z_{r,g_r(i,t)} + c_i \epsilon_i, \quad a_i^2 + \sum_r b_{ir}^2 + c_i^2 = 1. \quad (2)$$

Using $U_i = \Phi(Z_i)$ and $Y_{ik} = 1\{U_i \le F_{ik}\}$ preserves each individual CDF and gives ordered horizon outcomes. Shared draws persist across streamed entity chunks. Missing memberships transfer their squared loading to

private noise. Identities index incidence; no identity embedding or future assignment is used. The construction handles simultaneous job and machine membership without a dense entity covariance matrix.

Independent sampling uses only private noise. Global models learn state-specific global/private weights; typed models add one weight per observed relation type. The initial study also includes context-dependent weight offsets and an exponential common-shock family. All weights are nonnegative, with at least .02 private variance in the initial study and .10 in the new-source grid. These families cannot express arbitrary negative association. The initial study fits them by composite binary pair likelihood using earlier out-of-sample individual forecasts, random fitting horizon pairs, and January or original-development checkpoint selection. Different calibration treatments receive separate dependence fits in the first workflow replication; retrospective BPI 2017 diagnostics retain original weights to isolate marginal correction. The three-source extension instead fits small variance grids by count CRPS on its dependence window. A matched Student-t control divides Z_i by one shared $\sqrt{Q/\nu}$, $Q \sim \chi^2_\nu$, and uses the Student-t CDF to preserve uniform marginals. Its global control already contains the same shared scale, so additional gains cannot be attributed solely to adding heavy-tailed common variation.

The initial study permutes noise-sharing incidence within current state and half-unit last-horizon logit-risk bins, retaining true evaluation groups. This preserves individual probabilities, fitted weights and group sizes within strata. Additional Borg controls remove job or machine variance and transfer it to private noise. These sampler interventions assess predictive grouping under a fitted model, not causal effects of reassigning actual tasks or people. New-source coupling shuffles preserve known-membership masks and group sizes without risk stratification; neural shuffles preserve group sizes but may move missing-membership positions. These are distinct controls, not exact replicas of the original risk-stratified intervention.

3.2 Delayed calibration

A shared logit intercept preserves horizon order:

$$F_{ik}^{(\beta)} = \operatorname{logit}^{-1}\{\operatorname{logit}(F_{ik}) + \beta\}. \tag{3}$$

We fit $\beta \in [-2, 2]$ by equal-origin, equal-horizon Bernoulli loss with $10^{-4}\beta^2$ regularization. The static control uses an earlier calibration window. Rolling correction uses origins in the previous 60 days whose entire 14-day outcome window has matured; fewer than four eligible origins triggers an expanding-history fallback. Updated labels enter only after maturity.

A retrospective failure motivates an explicit version control: calibration residuals from earlier separately fitted predictors need not describe the current predictor's bias. We therefore compare that transfer with history from the same fitted predictor. This is a practical calibration control, not a new online-learning algorithm. Static calibration data also selected the original neural checkpoint/temperature, so its own fit score is not an independent validation estimate.

3.3 Direct counts and discrete reconciliation

For group A, average the individual first-exit category probabilities to obtain $\bar{\pi}_A$. A conditional Dirichlet–multinomial (DM) baseline uses

$$N_A \sim \mathrm{DM}(|A|, \kappa q_A), \qquad q_A = \operatorname{softmax}(\log \bar{\pi}_A + \delta). \tag{4}$$

We fit one concentration and optionally four category offsets (the fifth is zero) per projection type, with equal-origin and equal-within-origin-group likelihood weights. Offsets have fixed weak shrinkage. Concentration-only and bias-adjusted candidates are selected on development count CRPS. Training uses complete out-of-sample cohorts, never sampled partial group counts. Every group draw is bounded, integer-valued and ordered across horizons. Separately drawn overlapping projections need not correspond to a common entity realization.

For the retrospective original-log analyses we also fit DM on the more recent development forecasts, matching the fitting window of historical residuals. Leave-one-origin-out development CRPS selects concentration-only versus bias-adjusted DM; all development origins then fit the final predictor. These selection folds can use later development origins and are not online validation. Exact beta-binomial cumulative-count marginals supply CRPS and discrete quantiles for this control; joint-profile scores still use sampled DM trajectories. This stronger comparison avoids attributing an advantage to dependence when it comes from access to recent same-version errors.

For the workflow hierarchy, include every handler group, plus an unknown handler group, in a complete partition. Independently sampled bottom count vectors N_g are reweighted by the population forecast mass

$$w^{(m)} \;\propto\; p_{\mathrm{pop}}\left(\sum_g N_g^{(m)}\right). \tag{5}$$

This two-level conditioning adaptation [Zambon et al., 2024] uses 8,192 proposal draws and resamples 512 coherent group/population trajectories. We report effective sample sizes and verify its distribution against an enumerated small example. Bottom-up aggregation without conditioning is a separate control. Neither count method preserves individual CDFs. The implemented hierarchy is complete for workflow handlers; it is not a full crossed job–machine reconciliation implementation. Historical aggregate residual resampling remains a comparator in the original two logs.

4 Evaluation

We score current-state population counts and observed relation groups of at least five entities. Fair ensemble CRPS [Ferro, 2014] is divided by current group size, then averaged equally over groups within projection, horizons, projection types and origins. This weighting is a benchmark convention, not an inferred organizational utility. Coverage and width of 90% intervals are reported separately by projection.

For remaining current-cohort count $R = |A| - C$, the reserve proxy chooses forecast quantile b at $r/(r+1)$ and scores

$$\ell_r(b, R) = \{r(R-b)_+ + (b-R)_+\}/|A|, \quad r \in \{3, 9\}. \tag{6}$$

It measures a specified quantile tradeoff, not money or realized staffing savings. CRPS integrates quantile losses [Gneiting and Raftery, 2007]; improving it does not guarantee improvement at this particular quantile.

For joint group risk, let $x_g = C_g/|g|$. We use an energy score with $d(x, y) = \sqrt{G^{-1}\sum_g (x_g - y_g)^2}$ and a power-.5 variogram score [Scheuerer and Hamill, 2015]. The latter averages 2,048 outcome-independent ordered group pairs; subtracting the ensemble variance divided by draw count removes finite-ensemble bias of its squared moment error. Its estimate can be slightly negative. We additionally score the distribution of

$$D_k = \max_g \frac{\mu_{gk} - C_{gk}}{\sqrt{1 + \sum_{i \in g} F_{ik}(1 - F_{ik})}}, \qquad \mu_{gk} = \sum_{i \in g} F_{ik}. \tag{7}$$

This is the largest standardized slow-progress deviation relative to a fixed reference forecast. Every compared method uses the same reference F, including recalibrated methods. It measures forecast-relative group risk, not an observed capacity threshold or an organizational SLA.

The initial studies average three random seeds within each origin before paired circular moving-block resampling (10,000 replicates). Borg blocks span one/two days; BPI 2017 blocks span 14/28 days; the fresh BPI 2012 analysis prespecifies seven days with fourteen-day sensitivity. Short traces provide few blocks, so intervals describe these observed periods and offer limited evidence for other operating regimes. Cases and random seeds are not independent temporal replications. The three-source extension uses 2,000 replicates and 1/3/7-day blocks. Seven days is the entire Azure confirmation period, making that particular circular-bootstrap interval degenerate, not strong evidence of certainty. All intervals are descriptive, unadjusted for multiple comparisons.

5 Data and evidence stages

The sources span two compute providers and several workflow/incident releases (Table 1). Their horizons and target semantics differ; scores are compared within source, not pooled into one cross-domain leaderboard. All features and memberships use recorded events at or before the origin. Entire outcome windows must mature before their partition ends. Historical extracts do not establish complete live populations or ingestion availability.

The initial Borg/BPI 2017 study froze 264 assets. Its outcomes motivated retrospective direct-count, calibration, and joint-risk controls. BPI 2012 then froze 82 assets before February scoring. The new extension freezes 165 assets for Azure days 22–28.75, UCI May 1–28, 2016, and BPI 2014 January 1–28, 2014. The latter adds 84 origins without retuning on their scores. These are internal holdout freezes, not externally preregistered studies.

New-source model fitting, marginal calibration, dependence fitting, development validation, and confirmation occupy distinct chronological windows. Three-seed neural/boosted-tree CDF ensembles are selected by earlier calibration Brier; copulas and direct counts use the same dependence window. Controls include feature-matched temporal HGB, node-only and global-set networks, typed message passing, temporal attention, and relation-specific/shuffled graphs. Gaussian and Student-t couplings retain the selected marginal probabilities. Complete windows, feature exclusions, model budgets, and source-selection disclosures appear in Appendix A.

Azure retains all released VMs in a deterministic 5% sample of subscriptions, not a prefix of arbitrary VM rows. Google 2019 is audited on one shard only; incomplete trajectories prevent a valid predictive benchmark. BPI change snapshots lack reliable attribute-version histories. Citi Bike and two airline geographies are exploratory pilots, not additional confirmation sources.

6 Results

6.1 Incident relations help, but direct counts remain competitive

The new frozen periods contain 426,610 repeated entity–origin queries at 84 origins. On UCI incidents, Gaussian typed factors improve count-fraction CRPS from .04457 to .04311 relative to a global factor, a 3.3% reduction (Table 2). The three-day-block paired difference is $-.00146$, with interval $[-.00195, -.00098]$; the sign survives the one- and seven-day sensitivities (Figure 1). Shuffled incidence scores .04641. Thus, the gain is not reproduced by simply sharing noise among arbitrary groups of the same sizes.

Matched Student-t factors improve .04444 to .04284, a 3.6% typed gain; both use 16 degrees of freedom. Unlike the typed Gaussian's .847 coverage, typed Student-t reaches .922 and reduces reserve loss from .09042 to .08472. Its peak-deviation CRPS is 1.054 versus 1.381 for the global Student-t. However, bias-DM reaches .04277

Table 1: Six scored sources. An origin is a forecasting time, not an independent organizational replication. New-period experiments are selected after development pilots; their models are fixed before target preparation and scoring.

Source	Current-cohort target	Observed relations	Evidence stage	Origins
Borg 2011	First pending/running state exit	Job, machine	Original frozen period	28
BPI 2017	First qualifying application milestone	Last handler	Original frozen period	61
BPI 2012	First approval/decline/cancellation	Last handler	First confirmation	28
Azure V2	VM deletion	Deployment, subscription	New-period confirmation	28
UCI incidents	First recorded resolution/closure	Handler, team	New-period confirmation	28
BPI 2014	First Closed activity after Open	Team	New-period confirmation	28

Table 2: Frozen new-period results: count-fraction CRPS and ratio-9 reserve loss, both multiplied by 1,000 (lower is better). Each source has 28 origins. All joint arms share its development-selected three-seed marginal predictor. Direct rows use exact count distributions. Last row: typed Student-t (UCI), subscription-only Gaussian (Azure), team-only Gaussian (BPI14), selected on development, never on confirmation.

Model	UCI CRPS	Loss	Azure CRPS	Loss	BPI14 CRPS	Loss
Independent	48.63	148.60	8.70	20.83	34.76	81.82
Gaussian global	44.57	92.39	7.60	14.20	34.15	85.89
Gaussian typed	43.11	90.42	7.60	14.20	34.08	86.14
Shuffled typed	46.41	117.50	7.60	14.20	34.22	86.48
Student-t global	44.44	91.18	7.58	14.17	34.41	87.91
Student-t typed	42.84	84.72	7.58	14.17	34.40	88.25
Direct DM (exact)	43.31	87.15	7.47	13.19	33.70	82.46
Bias-DM (exact)	42.77	86.02	7.44	13.11	36.11	85.51
Residual	46.57	122.43	9.76	32.56	39.11	94.63
Dev.-selected coherent	42.84	84.72	7.25	13.00	33.74	78.76

Table 3: New-period equal-origin integrated Brier $\times 1,000$ (lower is better), using three-seed mean CDFs. Selected beforehand: typed graph (UCI), temporal HGB (Azure), relational HGB (BPI14). Single-relation order: handler/team for UCI; deployment/subscription for Azure. BPI14 has one relation.

Marginal model	UCI	Azure	BPI14
HGB personal	69.04	6.15	72.84
HGB relations	72.70	6.95	72.86
HGB temporal	70.09	6.83	72.45
Node MLP	68.07	6.40	74.14
Global set	69.24	6.59	74.83
Typed graph	67.88	6.45	74.27
Shuffled graph	68.67	6.60	—
Relation 1 only	68.07	6.30	—
Relation 2 only	67.76	6.39	—
Temporal attention	66.77	6.20	72.87

count CRPS: the Student-t-minus-bias-DM difference is only .000063, with three-day interval $[-.00187, .00155]$. The experiment supports useful incidence, not a resolved advantage over the strongest direct-count alternative.

Handler and team relations are not interchangeable. Handler-only Gaussian CRPS is .04449, team-only .04367, and their combined fit .04311. Adding a global component separately gives .04382 and .04359. The best observed Gaussian count score is not the best reserve score; global-plus-team loss is .08456, below the combined typed model's .09042. These are all frozen comparators, not a new policy selected on the final period.

6.2 Azure rewards a specific relation, not the combined fit

The arbitrary-row Azure pilot suggested roughly 5% additional typed gain. After retaining complete released subscriptions, the combined coordinate fit instead selected a global variance of .7 and zero typed variance. It therefore has exactly the global model's confirmation CRPS .00760. This is a result of the bounded fitting procedure, not proof that the globally optimal typed family has no useful relation component.

Separately fitted relation-only controls expose that distinction. Subscription-only Gaussian was the best coherent model on development and reaches .00725 on confirmation, 4.7% below global; deployment-only reaches .00728. Subscription-only reserve loss falls from .01420 to .01300 and peak CRPS from 2.178 to 1.606. Its count score is also below exact DM (.00747) and bias-DM (.00744). A supplementary, post-confirmation descriptive three-day interval for its difference from global is $[-.000450, -.000266]$; the model itself was fixed before confirmation. Only seven evaluation days are available, and no optimizer was retuned after observing this result. Table 8 reports every relation weight and comparison,

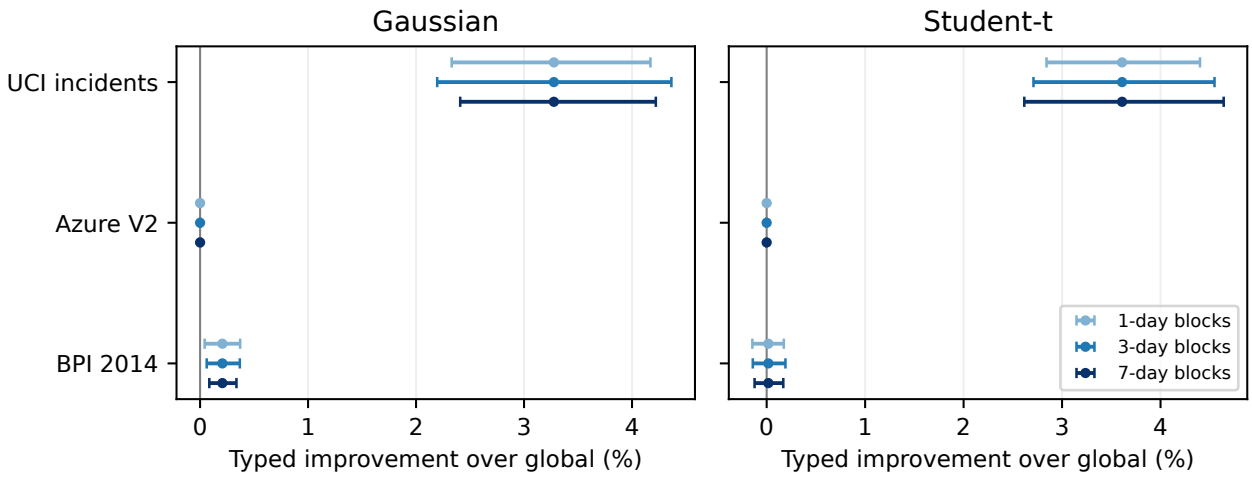


Figure 1: Primary typed-versus-global comparisons in the three new periods. Positive values favor typed incidence. Bars are descriptive paired 95% circular-block intervals, using 1/3/7-day blocks. Azure's combined typed and global fits are identical; these zero differences do not imply absence of all relation signal. Its separately fitted subscription-only control is reported in Table 8. A seven-day Azure block spans the whole trace, so that sensitivity is degenerate.

including the unsuccessful combined fit.

6.3 More complex models are not uniformly better

For BPI 2014, combined typed Gaussian improves global CRPS by only 0.2% (.03408 versus .03415), while matched Student-t has no resolved typed advantage. The development-selected team-only Gaussian reaches .03374 and .07876 reserve loss, versus direct DM's .03370 and .08246. Bias-DM was strong on development but deteriorates to .03611 on confirmation. Marginal correction and added tail flexibility therefore require temporal validation, even when they improve a different source.

The three-seed marginal comparisons also reject a universal neural ranking (Table 3). UCI's fixed typed graph beats the global-set and shuffled-graph controls, with Brier .06788 versus .06924 and .06868; temporal attention has the lowest observed score, .06677. Azure's personal HGB scores .00615, below its development-selected temporal HGB's .00683 and typed graph's .00645. BPI 2014 temporal HGB scores .07245 versus typed graph .07427. Joint results retain the predictor selected before confirmation, rather than switching to these observed winners.

A bounded retrospective Borg pilot similarly finds HGB with relation summaries at .05796 Brier versus typed message passing .07369. A job-only graph selected on calibration deteriorates to .07623 on development validation. These sampled scores are not the full-population Borg estimates. In the separate 64-node UCI pilot, the adapted TACTiS copula reaches .03761 count CRPS after rank adjustment, versus global Gaussian .03732 and typed Student-t .03482 on the same subsets. Its single-seed, discrete-surrogate adaptation is insufficient for a general claim about the published TACTiS models.

Negative and deferred pilots. Citi Bike and both New York and eleven-airport airline pilots provide no incremental combined typed-factor improvement over their global controls. The multi-airport validation CRPS is .01385 for both global and combined typed factors, versus .01432 for independence. Citi Bike's station-only arm improves global CRPS by about 0.5% on its small sampled period, insufficient to motivate a larger experiment under the screening rule. We retain these exploratory results without opening further transportation holdouts. Google 2019's incomplete shard histories and unversioned BPI change snapshots remain feasibility audits, not failed predictive scores or extra confirmation logs. This screening and all new experiments occupy approximately 1.1 GB locally; the large Google release was not downloaded.

6.4 Earlier evidence: the independence gap is not the typed gain

The original frozen Borg typed Gaussian model reduces count-fraction CRPS from .03417 under independence to .02587, a 24.3% improvement (Table 4). Ratio-9 reserve loss falls from .13106 to .05280. The global factor already reaches .02646 CRPS: the additional typed gain is 2.2%, not 24.3%. The paired typed-minus-global CRPS interval is $[-.00105, -.00018]$ with one-day blocks. Removing job variance increases job-level CRPS from .01451 to .01922; removing machine variance changes machine CRPS from .02507 to .02525. The strongest demonstrated typed signal is therefore job grouping, with a

Table 4: Count forecasts on original final periods and fresh BPI 2012 confirmation. Lower CRPS and loss are better; nominal coverage is .90. The three factor rows retain the original individual CDFs. Revised controls on Borg/BPI 2017 are retrospective. DM development-fit scores are exact; sampled methods use 512 draws and three seeds. A dash denotes an unimplemented comparison.

	Borg			BPI 2017			BPI 2012 (fresh)		
Method	CRPS	Cov.	Loss_9	CRPS	Cov.	Loss_9	CRPS	Cov.	Loss_9
Independent	0.0342	0.6594	0.1311	0.0501	0.5356	0.1177	0.0716	0.4745	0.1386
Global factor	0.0265	0.9797	0.0551	0.0460	0.8897	0.1290	0.0625	0.8562	0.1669
Typed factors	0.0259	0.9059	0.0528	0.0465	0.7935	0.1222	0.0653	0.6655	0.1554
DM, earlier fit	0.0287	0.9342	0.0744	0.0450	0.9002	0.1287	0.0389	0.9145	0.1125
DM, development fit	0.0285	0.9440	0.0693	0.0408	0.9298	0.1140	—	—	—
Bottom-up DM	—	—	—	0.0424	0.7749	0.1326	0.0389	0.8289	0.1176
Conditioned DM	—	—	—	0.0418	0.7617	0.1319	0.0390	0.8125	0.1198
Historical residuals	0.0309	0.8746	0.0840	0.0439	0.8311	0.1258	—	—	—

much smaller machine contribution.

Stronger direct controls do not remove Borg's count result. The development-fitted DM model reaches .02853 CRPS and .06931 reserve loss. Typed-minus-DM CRPS is $-.00266$, with one-day block interval $[-.00366, -.00169]$. The earlier-OOF DM and historical residual controls are weaker. Nevertheless, macro coverage is insufficient evidence of uniform calibration: original typed population coverage is about .763 while job and machine coverage are about .978 and .976. A macro value near .90 hides this imbalance.

BPI 2017 gives a different ranking. The original global model improves CRPS by 8.3% over independence but increases the reserve loss by 9.6%. Development-fitted bias-DM reaches .04077 CRPS and .11396 loss, outperforming the original global forecast on both. Its CRPS difference is $-.00518$ with 14-day block interval $[-.00983, -.00096]$. Conditioning reconciles handler/population trajectories but does not improve on the unreconciled direct count model; its population and group tradeoffs must be inspected separately. The early-OOF DM is weaker than its same-version development fit, showing why comparator training windows matter.

6.5 Calibration changes the operational comparison

The first retrospective correction transferred June–August OOF errors to the current BPI 2017 predictor. It made October global-model CRPS worse, from .04102 to .05422. Those OOF forecasts came from different fitted predictors. Using earlier September forecasts from the *same* predictor reverses that failure: delayed correction gives October CRPS .03845. The later-period global score improves from .04589 to .04072, with paired difference $-.00517$ and 14-day interval $[-.00859, -.00192]$. Its fourteen-day mean resolution forecast rises from .525 to .578, against observed .579. These are controlled follow-ups, not untouched final estimates.

The corrected global model and development-fitted direct DM have almost identical count CRPS: difference $-.00005$, interval $[-.00134, .00120]$. Correction therefore does not establish superiority over a strong direct alternative. Nor is the decision effect uniform. Corrected global reserve loss falls from .12899 to .11816, whereas corrected independence worsens from .11761 to .14405. The same improved marginal means produce different reserve quantiles under different joint laws (Table 5). Identity rows here use a common diagnostic random stream, so they differ slightly from the independently generated original frozen table.

6.6 The first workflow confirmation favors direct counts

The February BPI 2012 confirmation was evaluated once under the separately frozen protocol. The development-selected identity/global model improves CRPS by 12.7% over independence (.06253 versus .07162), but worsens reserve loss by 20.4% (.16694 versus .13864). Its CRPS difference is $-.00909$, with seven-day block interval $[-.01081, -.00721]$. Typed factors are worse than the global control by .00274 CRPS, interval $[.00187, .00357]$.

The development-selected bias-DM is substantially stronger: .03893 CRPS, .9145 coverage and .11246 reserve loss. Relative to the selected Gaussian, CRPS improves 37.7% and reserve loss 32.6%; the CRPS-difference interval is $[-.03183, -.01436]$. Bottom-up and conditioned bias-DM have similar count CRPS but less favorable coverage and reserve loss. Concentration-only DM is much weaker, implicating mean correction rather than dispersion alone.

This confirmation is a demanding extrapolation for the individual predictor, which had only 4,854 training case–origin queries before November. Its fourteen-day February mean is .468 versus observed .617. Static correction raises it to .575 and rolling correction to .532; both improve aggregate CRPS, but neither beats bias-DM. Static correction also outperforms rolling correction on count CRPS in this period. These outcomes confirm the need for strong bias/direct-count controls, not universal success of an online update or a conclusion about

the best possible individual model for this source. Only four seven-day or two fourteen-day blocks are available, limiting the strength of temporal inference.

6.7 Joint group risk changes the ranking

Table 6 evaluates full group profiles and the largest standardized slow-progress deviation. In BPI 2017, recent direct DM improves count CRPS over the original global model, but its peak-deviation CRPS is worse (.53433 versus .46081). The paired increase is .07351 with 14-day interval [.00348, .14454]. Thus access to better individual group-count distributions does not establish better prediction of their joint extreme. Delayed correction plus global dependence reaches .43633 peak CRPS and .08803 profile energy; typed correction offers no clear general advantage.

Borg's global and typed factors substantially improve peak-deviation scores over independence, primarily through jobs. Direct DM is also competitive on joint profiles and peaks, despite weaker count CRPS. In fresh BPI 2012, bias-DM wins both count and peak-deviation comparisons against the selected global model (.39031 versus .47902 peak CRPS). There is no universal coherence-versus-accuracy ranking. Full variogram results and projection-level scores are released rather than selecting only a favorable multivariate metric.

An earlier simultaneous-alarm endpoint required at least 20% of groups to fall more than two reference standard deviations behind. It was absent in Borg and occurred only once across BPI 2017 final horizon–origin outcomes. That endpoint is uninformative here and was not promoted as a successful warning task. Threshold-free profile and peak scores were specified before opening the fresh period.

7 Discussion

An evidence requirement, not a universal winning architecture. The benchmark exposes distinctions that an individual-event leaderboard cannot resolve. A shared factor can improve collective dispersion without changing a single individual probability. Typed incidence can then add a smaller, projection-specific gain. Correcting a common bias or forecasting counts directly can change the ranking again. These are empirical reasons to include global, direct-count and calibration controls before attributing collective accuracy to a temporal graph architecture.

Which output is needed? If the application asks only for a few fixed group counts, direct count models are strong and economical candidates. Entity-level factor scenarios supply a different capability: the same realization can be projected onto overlapping jobs and machines or an alternative grouping without fitting another count model. A complete handler hierarchy can also be reconciled coherently. Neither capability guarantees better prediction. In particular, separate good group marginals need not give the right extreme across groups, and a globally coherent scenario can still have biased means or incorrect tails. The joint profile and peak-deviation scores therefore complement count CRPS.

Calibration and dependence interact. Replacing biased individual probabilities changes both expected group counts and the quantile needed by a reserve proxy. A coupling that improved CRPS before correction need not have the same decision effect afterward. The same-version requirement matters because a sequence of chronological OOF predictors is not one stationary calibration object. Nevertheless, delayed updates can lag a regime change, and the fresh replication does not justify claiming that rolling correction always beats a fixed earlier map. A factor estimated from forecast residuals may absorb misspecification; its value is predictive rather than evidence of an identified common cause.

Limits of generalization. The evidence covers two compute providers, two loan-workflow releases, and two incident sources, but only seven final days per compute trace and 28 origins per confirmation source. Four weekly blocks in a monthly incident period are weak support for long-run uncertainty; an entire-trace block on Azure provides no meaningful interval variation. Repeated cases, millions of tasks and Monte Carlo seeds do not multiply independent temporal replications. Retrospective follow-ups are separated from their motivating outcomes. Development-based source promotion and a shared adapter introduce selection and implementation dependencies despite separate frozen periods. This should not be interpreted as broad organizational replication. No live operator study, policy deployment, measured cost saving or resource-capacity ground truth is available.

The benchmark conditions on the currently active cohort and excludes future arrivals, subsequent exits/returns and membership evolution. Last handler is an observed association, not exclusive ownership. Positive factor families cannot represent general negative dependence. A full crossed job–machine probabilistic reconciliation baseline, full TACTiS/TACTiS-2 models, graph point-process generators, and complete business-process simulation remain unimplemented comparisons. The small adapted attentional-copula pilot does not close that gap. The paper makes no superiority claim over those systems. Fixed-width neural controls, bounded history, and finite training budgets are useful comparisons, not an exhaustive architecture search. A typed copula gain is not evidence that a GNN is necessary: its shared probabilities may come from a tree. Conversely,

improved graph marginals do not by themselves identify the required joint outcome law.

Implications for scientific dataset generation. A reusable scenario source should declare the conditioning population, observable history and outcome contract before generating records. It should also test both individual and collective statistics under meaningful group projections. Our adapters, score suite and provenance checks provide a bounded foundation for that purpose. They are not a full synthetic-data factory and have not been validated by training downstream models on synthetic data. Extending them requires arrivals, repeated transitions and separate utility and privacy studies. Original public-source license and attribution terms remain applicable.

8 Conclusion

Across six operational logs, collective forecasting requires evidence beyond individual-event accuracy. Typed relations add measurable information for some job, subscription, and incident groups, but a global factor explains much of the independence gap, and strong direct-count or calibration controls can erase an apparent advantage. Temporal attention and message passing are useful comparators, not universal replacements for feature-matched trees. Reliable evaluation must separate marginal prediction, relation-specific dependence, and the particular collective decision being scored. Timestamp-bounded adapters, frozen confirmation assets, negative pilots, and projection-level scores make those distinctions reproducible while leaving live deployment and broader temporal replication open.

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A Detailed source and protocol definitions

A.1 Timestamp-bounded adapters

Borg. All 500 task-event shards of Google ClusterData2011-2 supply the compute application [Google, 2011]. We replay pending/running states in timestamp and source order, retaining unknown preceding age for time-zero records. The target is first state change at 1/5/15/60 minutes, including failures, evictions and kills as well as completion. Current job and known running-machine memberships define the graph. The evaluated final prefix contains 144.6 million events; 28 six-hour origins span elapsed days 22–28.75, with 3,760,001 task–origin queries and 105,077–171,603 active tasks per origin. Those repeated queries are not independent replications.

BPI 2017. The application workflow [van Dongen, 2017] uses first `A_Pending`, `A_Denied` or `A_Cancelled` after creation at 1/3/7/14 days. This is a documented application milestone, not repayment or necessarily the final event. Last observed workflow handling supplies the relation. We exclude retrospective trace attributes, eventual offer outcomes and future assignments. The original final period comprises 61 daily noon UTC origins during November–December 2016, with 114,051 case–origin queries and 1,484–2,189 cases per cohort.

BPI 2012. A fresh workflow release [van Dongen, 2012] supplies an analogous target: first `A_APPROVED`, `A_DECLINED` or `A_CANCELLED` after `A_SUBMITTED`. We normalize these four activity names only to reuse the replay adapter, retaining their original target definition. All other activity names remain distinct; workflow lifecycle values are normalized to lower case. Missing handlers share no entity-level factor. This is additional temporal/source evidence, not a claim about another independent bank. February 1–28, 2012 provides 28 final daily origins, with 23,749 case–origin queries, 793–889 cases per cohort, and follow-up through March 14 00:00 UTC. Activity vocabulary is derived from the individual fit prefix only.

In every source, observed events through the origin build features and memberships; outcomes are attached separately. Entire longest-horizon windows must be observable. Prefix state carries across date partitions without revealing future events. Recorded event time is the availability proxy; the archives do not establish live ingestion latency or exclusive resource ownership. The municipal BPI 2015 candidate was deferred after a development-source audit did not establish a sufficiently reliable closure definition across its activity labels. An Alibaba 2018 task-table candidate was also deferred: absent end timestamps dominated some reconstructed active cohorts, and

their interpretation as continued running was unverified. Neither candidate contributed predictive results.

A.2 Initial frozen evaluation

The initial study compares HGB personal/global/relational arms and temporal Mixer, GRU and tabular MLP ensembles. Borg uses 32 base features plus a 144-event-count extension; BPI 2017 uses 103 features. Temporal models use the last 16 events per task/job/machine or case/handler scope, width 64 and a five-category first-exit head. Selection uses equal-origin mean Brier over the four horizons, including monotone calibration variants and three-seed neural means; no best-seed selection is permitted.

Borg individual training/calibration/development end at days 18/20/22; dependence uses forward-fold predictions in days 8–10, 12–14 and 16–18, each produced by earlier fitted and calibrated predictors. BPI 2017 uses training before September, September calibration and October development, with June/July/August forward folds for dependence. Selected individual models are HGB global for pending tasks, HGB relational for running tasks, and the temperature-calibrated Mixer ensemble for BPI 2017. Gaussian typed and global models are selected for Borg and BPI respectively.

An immutable 264-file manifest froze code, models and evaluation choices before those final periods were opened. This was a development-informed holdout freeze, not external preregistration. A lossless Borg timestamp-type repair after a replay failure retained all original data and changed no timestamp value or frozen model. Original scientific results and frozen assets remain preserved; obsolete manuscript copies are not part of the current paper.

A.3 Retrospective diagnostics and fresh confirmation

The initial outcomes motivated the direct DM controls, delayed calibration, factor removals and joint-profile scores. Their analyses on the original final periods are therefore *retrospective*. Direct count training uses 24 complete Borg six-hour OOF cohorts and 50 complete BPI 2017 OOF cohorts. The stronger development-fitted DM control instead uses the eight Borg or 17 BPI 2017 current-version development origins, matching the historical residual fitting period. Both fitting regimes are retained. The original residual comparator fits on development residuals, with leave-one-origin-out development scoring and three sampling seeds.

For fresh confirmation, BPI 2012 individual fitting ends before November 1, 2011; November calibrates the individual model, December fits dependence and direct counts, and January selects configurations. The global-feature HGB with identity correction wins individual Brier. Identity/global wins Gaussian development count CRPS; bias-DM wins the direct comparison; bias-DM conditioning wins the conditioning comparison. Static and rolling correction remain prespecified controls, with dependence fitted separately for each treatment. All six Gaussian fits per seed use only development data.

Before retaining any February outcomes, a second immutable manifest froze 82 assets, the primary comparisons, metrics, origins and bootstrap settings. The raw parser had previously discarded every event at or after February 1 before retaining its activity or outcome. Final predictors remain fixed; only the explicitly prespecified rolling correction may use earlier February outcomes after their 14-day maturity delay. January 19–31 origins provide cross-boundary calibration history but are not final score origins. We retain all prespecified outcomes irrespective of which method wins.

A.4 Cross-domain extension and stronger graph controls

We next screened additional public sources with bounded local storage, retaining unsuccessful and unsuitable candidates. Azure V2 [Microsoft Azure, 2019] supplies a second compute provider. A first 49,231-row schema/prediction pilot was followed by a sample retaining *all released VMs* in subscriptions selected by a fixed SHA-256 rule (first 32 bits modulo 20 equal zero). The sample is complete within those released subscriptions, not the provider's full production population. The target is VM deletion at 5/30/120/360 minutes; observed deployment and subscription memberships supply two relations. Lifetime CPU summaries, usage-derived VM categories, and deletion times are excluded from features. Time-zero creations give an observed age lower bound; timestamp coarsening and the publisher's workload sampling limit interpretation. The interval adapter emits creation events only, so Azure temporal tokens summarize prior creations, not a full stream of VM deletions or CPU measurements.

The UCI incident log [Amaral et al., 2018] supplies handler and team relations. We predict the first *recorded* Resolved/Closed update after the first observed nonterminal update at 1/6/24/72 hours. This is an event-log milestone, not an estimate of an unobserved physical repair time. Resolution timestamps, closure codes, eventual resolvers, and SLA outcomes repeated in earlier rows are excluded. BPI 2014 [van Dongen, 2014] supplies timestamped incident activities and team membership; its target is the first Closed activity after an observed Open. We do not reconstruct live arrival populations from either historical case extract. Event timestamps proxy information availability. Naive incident timestamps are interpreted on a UTC clock for reproducible partitioning; the archives do not verify operational timezone or ingestion latency. This convention does not recover daylight-saving transitions.

All sources have distinct individual-fit, calibration,

dependence-fit, and development-validation windows. Entire longest-horizon labels must mature strictly before each window ends. Azure uses day boundaries 0/10/13/16/20; UCI uses March 1/22, April 1/15, and May 1, 2016; BPI 2014 uses October 1, November 1/15, December 1, 2013, and January 1, 2014. Prefix state carries across boundaries. New-period confirmation fixes 28 origins per source: Azure days 22–28.75, UCI May 1–28, and BPI 2014 January 1–28. An executable asset manifest freezes all fitted models and comparisons before these targets are prepared or scored. Earlier schema audits inspected whole-file state totals and time ranges; this is an internal prediction-holdout freeze, not external preregistration or a claim that the raw files were never opened.

Matched individual models. Controls include personal-feature HGB, HGB with current relation summaries, and HGB with the same observed event-history tokens available to temporal attention. Neural controls use width 48: a node MLP, a global set model, two rounds of typed entity–group mean message passing, and temporal attention over the last eight observed relation events followed by message passing. These are independent survival-target adaptations, not reproductions of TGN, TGAT, or DyGFormer benchmark systems [Rossi et al., 2020, Xu et al., 2020, Yu et al., 2023]. Each network uses a five-category first-exit head. UCI trains for 30 epochs; Azure and BPI 2014 for 20, with AdamW learning rate .001 and weight decay .001. HGB uses 100 boosting iterations, 15 leaves and L2 penalty 10. Checkpoints and temperature are chosen on the earlier calibration window; three-seed mean CDFs determine the selected architecture. Relation-specific and size-preserving shuffled graph controls keep the input feature set and training budget fixed. A separate bounded Borg pilot shares the existing 176 engineered features across models, testing additional message passing beyond those summaries; its sampled-node scores are not population estimates.

Joint controls. Gaussian factors include independent, global, each relation alone, global-plus-each-relation, combined typed, and shuffled-incidence controls. Small development-only variance grids and coordinate search fit count scores; the typed model may reduce to the global control. Coordinate fitting starts at the global fit and performs two sweeps over weights; it is not an exhaustive joint-grid optimum. Separately fitted relation-only arms are retained even when they beat that combined fit. Student-t copulas add a shared inverse-chi-square scale and compare global and typed variants with degrees of freedom chosen from 4/8/16. Each draw still uses one entity uniform across horizons. Concentration-only and bias-adjusted direct DM, exact beta-binomial count scores, and historical residual forecasts use the same dependence-fitting period. None is tuned on confirmation.

A small additional comparator uses the unchanged published TACTiS attentional copula module [Drouin et al., 2022], with a custom current-cohort encoder, randomized discrete probability-integral-transform targets, and rank-adjusted generated uniforms. All comparators in that pilot use the same 64-node subsets. It is explicitly a limited adaptation: neither the full time-series model nor an exact discrete likelihood reproduction, and not evidence of superiority over TACTiS or TACTiS-2 in their original tasks.

Exploratory fitting uses empirical-ensemble CRPS; confirmation uses the fair distinct-pair estimator with 512 draws and three sampling seeds. The peak reference scale in confirmation is $\sqrt{1+\sum_i F_i(1-F_i)}$, as in the original study. Development pilot peak scores used a max-with-one stabilization and are not numerically pooled with confirmation. Seeds are averaged within origin before 2,000 paired circular block replicates with 1/3/7-day blocks. The same frozen probabilities and reference risk quantities are used for every joint comparator.

Screened alternatives and limits. A Google 2019 [Google, n.d.] event-shard audit found approximately 83% of sampled entity histories contained one event; that shard cannot support a complete-trajectory forecast. BPI 2014 change snapshots lack version histories for planned times and affected configuration items; incident activities offer a better-supported relation replay. Citi Bike [Citi Bike, n.d.] and airline [Bureau of Transportation Statistics, 2024] pilots test transportation at small scale. Published trips are retrospectively filtered, and airline schedule-version availability is an explicit assumption. Their exploratory scores are not presented as live deployment results. Source selection followed development feasibility and performance, so reporting all screened candidates is part of the evidence.

B Implementation and reproduction details

Individual and dependence models. The original HGB runs use 250 iterations, 31 leaves, minimum leaf size 100, learning rate .05 and L2 regularization 1. Neural models use width 64, up to 30 epochs, AdamW at .001, gradient clipping at 5 and calibration-window early stopping with patience 5. Case/job/machine histories contain at most 16 prior events per scope; time lags are computed at the origin. All initial neural architectures have three-seed ensembles. The fresh HGB control uses 15 leaves and minimum leaf size 50 because its fit window is smaller; it does not repeat the full neural architecture search. This difference is a limitation of the fresh replication, not a claim of universal model optimality.

Pair sampling draws 1,000 uniform entity pairs per origin and, per relation, 1,000 uniformly selected group pairs and 1,000 group-size-weighted pairs. It never conditions on future labels. Static/global AdamW uses learning rate .02 and up to 50 epochs, with development pair-likelihood patience 8. The context model in the initial study uses two width-32 SiLU layers, bounded logit offsets and shrinkage .01. Gaussian probabilities use 32-point double-precision quadrature. Private variance is at least .02. The common-shock, context and complete individual-arm results remain in the original artifact; no consistent context-network advantage was established.

DM fits use $\log \kappa \in [-2, 12]$ and four offsets in $[-2, 2]$, with $10^{-3}\|\delta\|^2$ penalty. Two fixed concentration initializations are optimized with analytic gradients and a converged minimum is retained. A first finite-difference Borg fit stopped at a line-search failure before producing scores; its directory and source snapshot are preserved. The analytic-gradient correction changed the repeated BPI count CRPS by less than 4×10^{-6}. Exact cumulative beta-binomial CRPS is checked against an enumerated pairwise expectation. Reconciliation uses log-normalized importance weights; all fitting and sampling functions are independently checked on small known distributions. For the updated BPI 2017 reconciliation control, the importance-pool effective-sample fraction averages .801 (minimum .442). Across both prespecified fresh BPI 2012 DM variants it averages .869 (minimum .654). These checks do not indicate severe weight collapse in the evaluated pools; they do not establish exactness of the finite-pool approximation.

Historical residual comparator. For each development origin/group, store the four-dimensional residual $(C - \mu)/\sqrt{1 + \sum_i F_i(1 - F_i)}$. Each draw selects one common historical origin, then size-bin-matched group residual vectors. Counts are clipped, rounded and monotonized. Development scoring omits its own origin; final scoring uses all development residuals. It changes group means and does not enforce agreement between overlapping projections. Contrary to one caption in the original review PDF, its saved evaluator and result files use seeds 0, 1 and 2, not one seed. The correction affects reporting only.

Interval and Monte Carlo interpretation. Seed averaging occurs before temporal block resampling. The original 512-to-1,024-draw sensitivity changes macro CRPS by at most .000122 and preserves within-family rankings. This does not establish sampling precision for every newly added peak or variogram score; close differences are not treated as decisive. Fair variogram estimates use sampled group pairs and can be slightly negative after finite-ensemble correction. Main tables separate exact direct-count marginal scores from sampled distributions; exact discrete quantiles and finite-ensemble quantiles can differ slightly. Projection/horizon CSVs and paired block-size sensitivities accompany the reported macro values.

A post-confirmation extension check doubles scenario draws to 1,024 for seed 0, without retraining or reselecting models. UCI's typed/global advantage and Azure's subscription-only/global advantage persist, while BPI 2014's combined typed/global difference remains small. This checks numerical stability of count scores, not independent generalization or all tail metrics.

Computation and integrity. The experiments use two RTX A6000 GPUs, with CPU replay, HGB and direct-count fits. Original Borg evaluation of five controls over 28 origins requires about 245–251 seconds per 512-draw process under concurrent shared-device execution, with at most .67 GiB allocated PyTorch memory; it excludes replay, training and individual inference and is not an isolated latency benchmark. The original 264-entry, first-confirmation 82-entry, and new-extension 165-entry asset manifests are verified after evaluation. Original scientific code, results and frozen assets remain intact; obsolete manuscript copies were removed. New diagnostics and manuscript sources live in separate directories. The extension additionally checks that later source records do not alter earlier observed attributes, relation partitions or training encodings. The test suite covers causal replay, temporal eligibility, coherent sampling, known DM moments, exact reconciliation on a small enumerated distribution, and dependence-sensitive profile scores.

Data and access. Raw-source manifests retain checksums and publisher URLs. Google trace attribution follows its CC-BY-4.0 release; BPI datasets retain the 4TU General Terms of Use specified in their metadata. The artifact does not relicense third-party data or software. The reproduction package supplies executable analysis, fitted parameters and score evidence; large original logs are acquired through the documented source workflow. Author identities, permanent public artifact links and any venue-specific disclosure must be finalized by the researcher before submission. No external submission or release has been made during this study.

Table 5: BPI 2017 controlled same-version correction. Original Gaussian weights are fixed. These are retrospective diagnostics; improved mean calibration does not guarantee a better reserve quantile.

		October development		November–December	
Correction	Dependence	CRPS	Loss_9	CRPS	Loss_9
Identity	Independent	0.0433	0.0954	0.0501	0.1176
Identity	Global factor	0.0410	0.1195	0.0459	0.1290
Identity	Typed factors	0.0411	0.1088	0.0465	0.1222
Static	Independent	0.0401	0.0945	0.0469	0.1197
Static	Global factor	0.0386	0.1147	0.0435	0.1244
Static	Typed factors	0.0383	0.1039	0.0439	0.1188
Rolling 60D	Independent	0.0401	0.0947	0.0435	0.1441
Rolling 60D	Global factor	0.0385	0.1146	0.0407	0.1182
Rolling 60D	Typed factors	0.0382	0.1037	0.0408	0.1199

Table 6: Joint group-profile scores. ES is RMS-normalized energy score; Peak is CRPS of the maximum standardized slow-progress deviation. Scales differ across datasets. Original-period rows are retrospective; all fresh endpoints were frozen before opening February.

	Borg		BPI 2017		BPI 2012 (fresh)	
Method	ES	Peak	ES	Peak	ES	Peak
Independent	0.0597	15.7604	0.0957	0.4613	0.1058	0.5183
Global factor	0.0575	7.5434	0.0942	0.4608	0.1023	0.4790
Typed factors	0.0546	4.1726	0.0942	0.6686	0.1015	0.7362
DM, earlier fit	0.0543	3.9903	0.0923	0.9984	0.0859	0.3903
DM, development fit	0.0544	4.0844	0.0879	0.5343	—	—
Bottom-up DM	—	—	0.0879	0.5380	0.0861	0.3907
Conditioned DM	—	—	0.0879	0.5539	0.0860	0.3922
Rolling correction + global	—	—	0.0880	0.4363	0.0928	0.4882

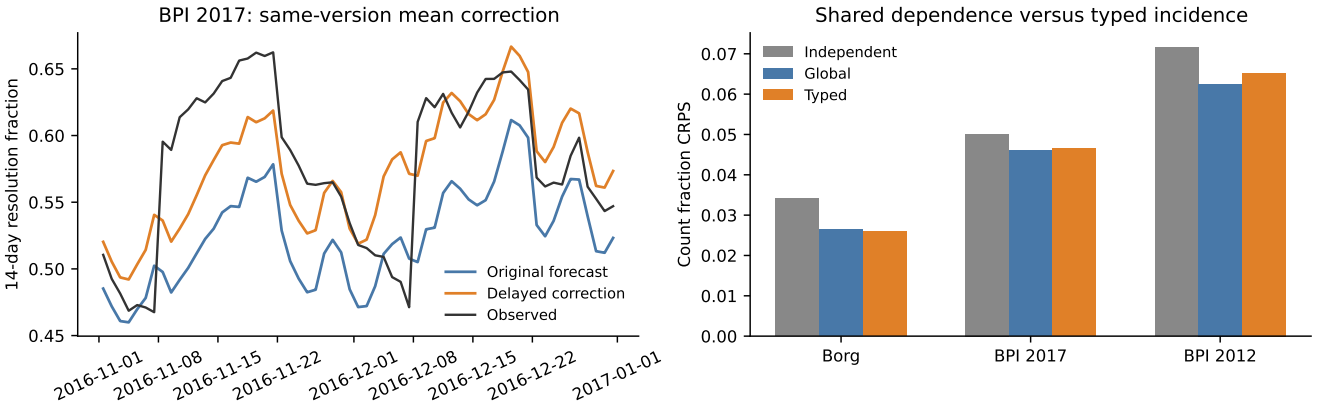


Figure 2: Retrospective BPI 2017 fourteen-day population forecasts under original and delayed same-version correction (left); matched-marginal factor comparisons on original final periods and fresh BPI 2012 (right). Global dependence captures much of the independence improvement; typed incidence adds a smaller, application-specific contribution.

Table 7: All-source confirmation distribution diagnostics. Coverage targets .90; width is in cohort fractions. Energy, fair variogram and peak-deviation CRPS are lower-is-better. Direct DM joint profiles are separately generated group trajectories, not coherent entity-level realizations.

Source	Model	Cov.	Width	Energy	Variogram	Peak
UCI incidents	Independent	0.669	0.144	0.06817	0.02670	1.791
UCI incidents	Gaussian global	0.908	0.236	0.06524	0.02689	1.396
UCI incidents	Gaussian typed	0.847	0.263	0.06433	0.02568	1.107
UCI incidents	Student-t global	0.937	0.247	0.06536	0.02692	1.381
UCI incidents	Student-t typed	0.922	0.277	0.06371	0.02563	1.054
UCI incidents	Direct DM	0.931	0.241	0.06411	0.02544	1.094
UCI incidents	Bias-DM	0.935	0.269	0.06366	0.02613	1.069
Azure V2	Independent	0.765	0.022	0.03744	0.01045	2.698
Azure V2	Gaussian global	0.985	0.041	0.03720	0.01018	2.178
Azure V2	Gaussian typed	0.985	0.041	0.03720	0.01018	2.178
Azure V2	Student-t global	0.981	0.040	0.03716	0.01023	2.165
Azure V2	Student-t typed	0.981	0.040	0.03716	0.01023	2.165
Azure V2	Direct DM	0.950	0.033	0.03446	0.01006	1.650
Azure V2	Bias-DM	0.948	0.031	0.03439	0.00986	1.629
BPI 2014	Independent	0.693	0.118	0.04777	0.01505	1.349
BPI 2014	Gaussian global	0.915	0.166	0.04736	0.01520	1.237
BPI 2014	Gaussian typed	0.927	0.177	0.04718	0.01506	1.211
BPI 2014	Student-t global	0.948	0.178	0.04776	0.01534	1.250
BPI 2014	Student-t typed	0.955	0.189	0.04761	0.01520	1.240
BPI 2014	Direct DM	0.915	0.166	0.04658	0.01488	1.174
BPI 2014	Bias-DM	0.848	0.155	0.04906	0.01503	1.411

Table 8: Frozen relation controls. Variance weights are global/handler/team (UCI), global/deployment/subscription (Azure), and global/team (BPI14). Only degrees of freedom change between matched Gaussian and Student-t arms; Student-t uses a shared scale. Scores are $\times 1,000$.

Source	Coupling	Weights	ν	CRPS	Loss
UCI incidents	G: global	0.15/0/0	—	44.57	92.39
UCI incidents	G: handler	0/0.5/0	—	44.49	105.93
UCI incidents	G: global handler	0.15/0.15/0	—	43.82	86.52
UCI incidents	G: team	0/0/0.3	—	43.67	95.06
UCI incidents	G: global team	0.15/0/0.15	—	43.59	84.56
UCI incidents	G: typed	0/0.3/0.3	—	43.11	90.42
UCI incidents	t: global	0.15/0/0	16	44.44	91.18
UCI incidents	t: typed	0/0.3/0.3	16	42.84	84.72
Azure V2	G: global	0.7/0/0	—	7.60	14.20
Azure V2	G: deployment	0/0.7/0	—	7.28	13.09
Azure V2	G: global deployment	0.7/0/0	—	7.60	14.20
Azure V2	G: subscription	0/0/0.7	—	7.25	13.00
Azure V2	G: global subscription	0.7/0/0	—	7.60	14.20
Azure V2	G: typed	0.7/0/0	—	7.60	14.20
Azure V2	t: global	0.7/0/0	4	7.58	14.17
Azure V2	t: typed	0.7/0/0	4	7.58	14.17
BPI 2014	G: global	0.05/0	—	34.15	85.89
BPI 2014	G: team	0/0.15	—	33.74	78.76
BPI 2014	G: global team	0.05/0.05	—	34.08	86.14
BPI 2014	G: typed	0.05/0.05	—	34.08	86.14
BPI 2014	t: global	0.05/0	16	34.41	87.91
BPI 2014	t: typed	0.05/0.05	16	34.40	88.25